

SYDNEY GRAMMAR SCHOOL



2017 Assessment Examination

FORM VI

MATHEMATICS EXTENSION 2

Thursday 18th May 2017

General Instructions

- Writing time — 2 hours
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 70 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eleven.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Reference sheet
- Candidature — 73 boys

Examiner

LRP

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

The ellipse $9x^2 + 16y^2 = 144$ has eccentricity $\frac{\sqrt{7}}{4}$. What are the coordinates of its foci? **1**

- (A) $S(0, \sqrt{7})$ and $S'(0, -\sqrt{7})$
- (B) $S(\sqrt{7}, 0)$ and $S'(-\sqrt{7}, 0)$
- (C) $S(4\sqrt{7}, 0)$ and $S'(-4\sqrt{7}, 0)$
- (D) $S(0, 4\sqrt{7})$ and $S'(0, -4\sqrt{7})$

QUESTION TWO

What is the remainder when $P(z) = 2z^3 - 3z^2 + 4z - 2$ is divided by $(z + i)$? **1**

- (A) $1 - 2i$
- (B) $1 - 6i$
- (C) $1 + 2i$
- (D) $1 + 6i$

QUESTION THREE

Every point on a certain conic is twice as far from the line $x = 4$ as from the point $(1, 0)$. **1**
What is a possible equation of the conic?

- (A) $\frac{x^2}{3} - \frac{y^2}{4} = 1$
- (B) $\frac{x^2}{4} - \frac{y^2}{3} = 1$
- (C) $\frac{x^2}{3} + \frac{y^2}{4} = 1$
- (D) $\frac{x^2}{4} + \frac{y^2}{3} = 1$

QUESTION FOUR

Two of the zeroes of the polynomial $P(x) = x^4 - 4x^3 + 9x^2 - 16x + 20$ are $a + ib$ and $2ib$, where a and b are real and $b \neq 0$. What is the value of a ? **1**

- (A) 2
- (B) -2
- (C) 4
- (D) -4

QUESTION FIVE

Which of the following is equivalent to $\int_a^b x^3 e^{2x^4} dx$, where a and b are real constants? **1**

- (A) $\int_{a^4}^{b^4} e^{2u} du$
- (B) $\frac{1}{8} \int_a^b e^u du$
- (C) $\frac{1}{4} \int_{a^4}^{b^4} e^{2u} du$
- (D) $\frac{1}{8} \int_{8a^3}^{8b^3} e^u du$

QUESTION SIX

Let x metres be the displacement of a particle of mass 1000 kilograms from the origin on a straight path. The particle experiences a constant propelling force of 10 000 newtons and a resistive force of magnitude $100v^2$ newtons, where v is the velocity of the particle at time t seconds. What is the equation of motion of the particle? **1**

- (A) $\ddot{x} = 10\,000 - 100v^2$
- (B) $\ddot{x} = 10 - 0.1v^2$
- (C) $\ddot{x} = 10\,000 - 0.1v^2$
- (D) $\ddot{x} = 10 - 100v^2$

QUESTION SEVEN

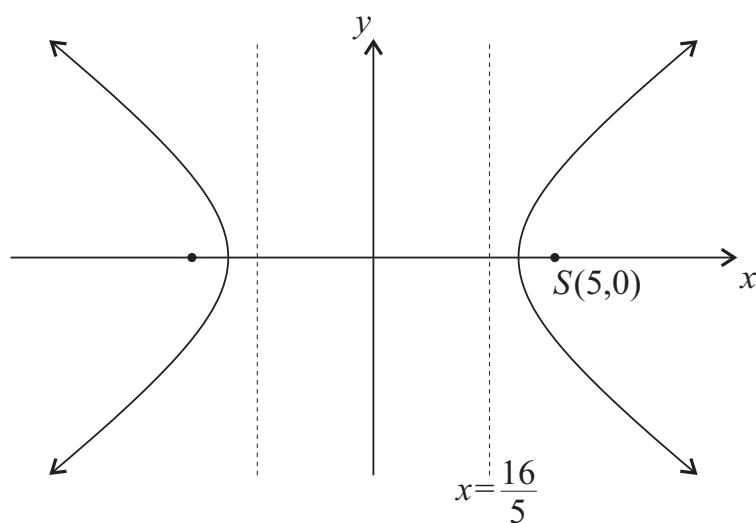
Let $x = \sin \theta - \cos \theta$ and $y = \frac{1}{2} \sin 2\theta$. What is the correct expression for $\frac{dy}{dx}$?

1

- (A) $\cos \theta - \sin \theta$
- (B) $\sec \theta + \operatorname{cosec} \theta$
- (C) $\sec \theta - \operatorname{cosec} \theta$
- (D) $\cos \theta + \sin \theta$

QUESTION EIGHT

1



A hyperbola centred at the origin has a focus at $S(5,0)$ and a directrix $x = \frac{16}{5}$.

What is the eccentricity of the hyperbola?

- (A) $\frac{4}{3}$
- (B) $\frac{25}{16}$
- (C) $\frac{5}{4}$
- (D) 4

QUESTION NINE

1

QUESTION TEN

Consider the relation $a^2x^2 + (1 - a^2)y^2 = b^2$, where a and b are non-zero real numbers.

1

Which of the following CANNOT be represented by the relation?

- (A) a circle
- (B) a parabola
- (C) a hyperbola
- (D) a pair of straight lines

End of Section I

Examination continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

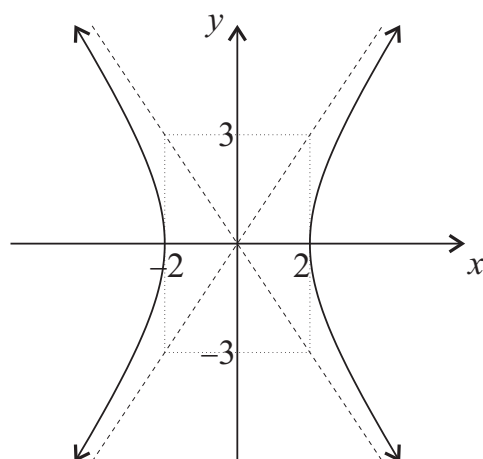
Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.

Marks

(a)



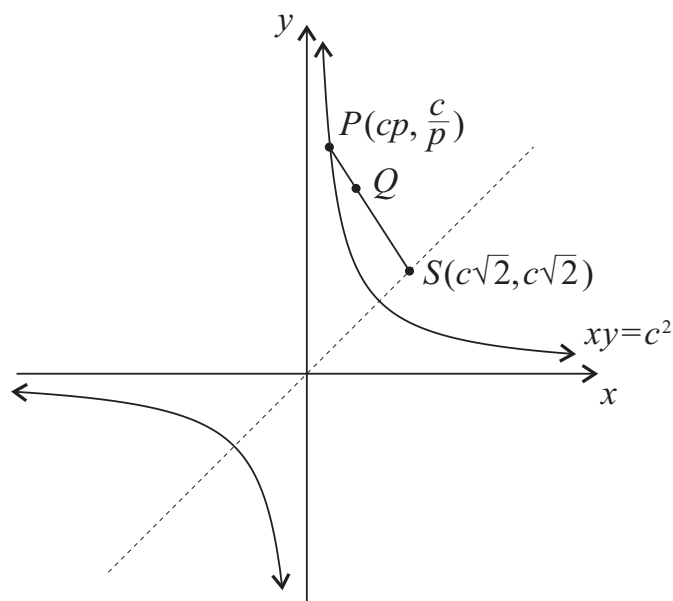
The diagram shows a hyperbola with asymptotes $y = \frac{3x}{2}$ and $y = -\frac{3x}{2}$.

- | | |
|--|----------|
| (i) Write an equation for the hyperbola. | 1 |
| (ii) Find the eccentricity of the hyperbola. | 1 |
| (iii) Write down the coordinates of both foci of the hyperbola. | 1 |
| (iv) Write down the equations of both directrices of the hyperbola. | 1 |
|
(b) Consider the polynomial $P(x) = 3x^3 - 10x^2 + 7x + 10$. | |
| (i) Given that one zero of $P(x)$ is $2 - i$, find the other two zeroes. | 2 |
| (ii) Hence express $P(x)$ as the product of a linear factor and a quadratic factor, both with real coefficients. | 2 |
|
(c) The polynomial equation $2x^3 - 9x^2 + 12x - 4 = 0$ has a double root at $x = \alpha$. | |
| (i) Find the value of α . | 2 |
| (ii) Find the remaining root. | 1 |

Examination continues next page ...

QUESTION ELEVEN (Continued)

(d)



The point $P\left(cp, \frac{c}{p}\right)$, where $p > 0$, lies on the rectangular hyperbola $xy = c^2$ with focus $S\left(c\sqrt{2}, c\sqrt{2}\right)$. The point Q divides the interval PS in the ratio $1 : 2$.

- (i) Show that the coordinates of Q are $\left(\frac{2cp + c\sqrt{2}}{3}, \frac{2c + cp\sqrt{2}}{3p}\right)$. **2**
- (ii) Find the Cartesian equation of the locus of Q as P varies. **2**

QUESTION TWELVE (15 marks) Use a separate writing booklet.**Marks**

- (a) When a polynomial $P(x)$ is divided by $(x-2)$ and $(x-3)$ the respective remainders are 4 and 3. Determine what the remainder must be when $P(x)$ is divided by $(x-2)(x-3)$. **3**

- (b) Barbara decides to go bungee jumping. This involves being tied to a bridge by an elastic cable of unstretched length d metres and falling vertically from rest from this point. After Barbara free falls d metres, she will be slowed down by the cable, which exerts a resistive force proportional to the distance greater than d that she has fallen.

If we take the origin at bridge level, x to be the distance fallen in metres and g to be the acceleration due to gravity in ms^{-2} , then Barbara's motion during her initial descent will be defined by:

$$\ddot{x} = \begin{cases} g & \text{when } x \leq d \\ g - k(x - d) & \text{when } x > d \end{cases}$$

Let Barbara's speed be $v \text{ ms}^{-1}$.

- (i) Find an expression for v^2 at the instant when Barbara first passes $x = d$. **2**
- (ii) Hence show that $v^2 = 2gx - k(x - d)^2$ for $x > d$. **2**
- (c) A ball is thrown vertically upwards with an initial velocity of $7\sqrt{6} \text{ ms}^{-1}$. It is subject to gravity and air resistance. The acceleration of the ball is given by $\ddot{x} = -(9.8 + 0.1v^2)$, where x metres is its vertical displacement from the point of launch and $v \text{ ms}^{-1}$ is its velocity at time t seconds.
- (i) Find an expression for time t as a function of velocity v . **3**
- (ii) Hence find the time taken for the ball to reach its maximum height. Give your answer correct to three significant figures. **1**
- (iii) Find an expression for vertical displacement x in terms of velocity v . **3**
- (iv) Hence find the maximum height reached. Give your answer in exact form. **1**

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.

Marks

- (a) The rise and fall in sea level due to tides can be modelled with simple harmonic motion. On a certain day, a channel is 8 metres deep at 7 am when it is low tide, and 14 metres deep at 2 pm when it is high tide.
- (i) Sketch a graph showing the depth of the water d at time t . Write an equation that models the depth of water d as a function of time t . Take the origin of time to correspond to the low tide at 7 am. **3**
 - (ii) A ship must sail down the channel at some time between 7 am and 9 pm. If the ship requires a water depth of at least 12 metres, between what times of day can the ship pass safely through? Give your answer correct to the nearest minute. **3**
- (b) The roots of $2x^3 - 9x^2 + 8x - 2 = 0$ are α , β and γ .
- (i) Find the value of $\alpha\beta\gamma$. **1**
 - (ii) Hence find a simplified cubic polynomial equation with integer coefficients that has roots $\frac{\alpha\beta}{\gamma}$, $\frac{\alpha\gamma}{\beta}$, and $\frac{\beta\gamma}{\alpha}$. **3**
- (c) The equation $x^3 - 3ax + b = 0$, with real constants $a > 0$ and $b \neq 0$, has three distinct real roots.
- (i) Find the stationary points of $y = x^3 - 3ax + b$ in terms of a and b and determine their nature. **3**
 - (ii) Hence show that $b^2 < 4a^3$, explaining your reasoning carefully. **2**

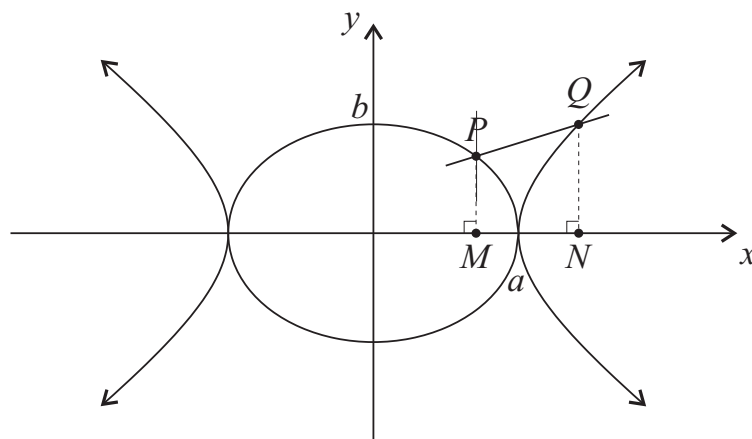
QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**

(a) Let $I_n = \int_0^1 \frac{1}{(x^2 + 1)^n} dx$ for any integer $n \geq 1$.

(i) Show that $I_{n+1} = \frac{1}{2n} [2^{-n} + (2n - 1) I_n]$. **4**

(ii) Hence evaluate I_3 . **2**

(b)



Distinct points $P(a \cos \theta, b \sin \theta)$ and $Q(a \sec \theta, b \tan \theta)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ respectively, as shown, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. The points M and N are the feet of the perpendiculars from P and Q respectively to the x -axis.

(i) The line PQ meets the x -axis at K . Show that $\frac{KM}{KN} = \cos \theta$. **1**

(ii) Hence find the coordinates of K . **2**

(iii) Show that the tangent to the ellipse at P has equation $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$ and deduce that it passes through N . **3**

(iv) The tangent to the hyperbola at Q has equation $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$ and passes through M . Do NOT prove this. Let T be the point of intersection of PN and QM .

(α) Show that T always lies on the same vertical line and state its equation. **1**

(β) Where on this line can T lie? Justify your answer. **1**

(γ) Suppose that θ is now in the second or third quadrant. Explain where T may lie. **1**

End of Section II

END OF EXAMINATION

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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question OneA ☐ B ☐ C ☐ D ☐**Question Two**A ☐ B ☐ C ☐ D ☐**Question Three**A ☐ B ☐ C ☐ D ☐**Question Four**A ☐ B ☐ C ☐ D ☐**Question Five**A ☐ B ☐ C ☐ D ☐**Question Six**A ☐ B ☐ C ☐ D ☐**Question Seven**A ☐ B ☐ C ☐ D ☐**Question Eight**A ☐ B ☐ C ☐ D ☐**Question Nine**A ☐ B ☐ C ☐ D ☐**Question Ten**A ☐ B ☐ C ☐ D ☐

EXTENSION 2 - SOLUTIONS

May Assessment 2017

Q1. $\frac{9x^2}{144} + \frac{16y^2}{144} = \frac{144}{144}$

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

$$a = 4, e = \frac{\sqrt{7}}{4} \therefore ae = \sqrt{7}$$

$$\therefore \text{Foci: } (\pm\sqrt{7}, 0)$$

B

Q2. $P(-i) = 2(-i)^3 - 3(-i)^2 + 4(-i) - 2$
 $= 2i + 3 - 4i - 2$
 $= 1 - 2i$

A

Q3. $e = \frac{1}{2} \therefore$ ellipse

Focus: $(1, 0)$ & vertical directrix

D

Q4. Roots must be $a+ib, a-ib, 2ib, -2ib$

$$\text{Sum of roots: } 2a = \frac{-(-4)}{1}$$

$$\therefore a = 2$$

A

Q5. $\int_a^b x^3 e^{2x^4} dx$

Let $u = x^4$
 $du = 4x^3 dx$

$$= \frac{1}{4} \int_a^b 4x^3 e^{2x^4} dx$$

$$= \frac{1}{4} \int_{a^4}^{b^4} e^{2u} du$$

x	a	b
u	a^4	b^4

C

Q6.

$$\begin{array}{ccc} \longleftarrow & \bullet & \longrightarrow \\ 100v^2 \text{ N} & & 10000 \text{ N} \end{array}$$

$$1000 \ddot{x} = 10000 - 100v^2$$

$$\ddot{x} = 10 - 0.1v^2$$

B

Q7. $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

$$= \frac{\cos 2\theta}{\cos \theta + \sin \theta}$$

$$= \frac{\cos^2 \theta - \sin^2 \theta}{\cos \theta + \sin \theta}$$

$$= \frac{(\cos \theta + \sin \theta)(\cos \theta - \sin \theta)}{\cos \theta + \sin \theta}$$

$$= \cos \theta - \sin \theta$$

A

Q8. $ae = 5$ — (1)

$$\frac{a}{e} = \frac{16}{5} \quad \text{--- (2)}$$

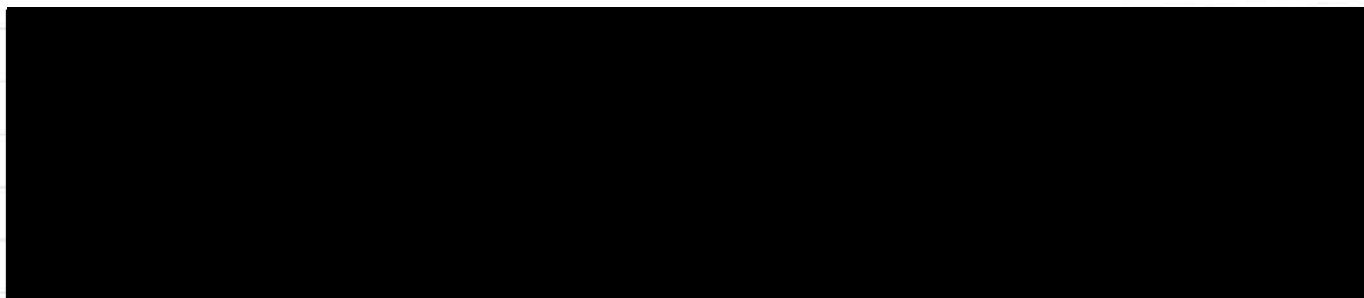
$$\textcircled{1} \times \textcircled{2}: \quad a^2 = 16$$

$$a = 4$$

$$\therefore e = \frac{5}{4}$$

C

Q9.



Q10. $a^2x^2 + (1-a^2)y^2 = b^2$

circle ✓ $a^2 = 1 - a^2$
 $\therefore a^2 = \frac{1}{2}$

hyperbola ✓ $1 - a^2 < 0$
 $a^2 > 1$

straight lines ✓ $a^2 = 0 \rightarrow y = \pm b$ (but a is non-zero anyway...)
 $a^2 = 1 \rightarrow x = \pm b$

$\therefore$ parabola

B

QUESTION 11:

a) i) $a = 2$
 $b = 3$

$$\frac{x^2}{4} - \frac{y^2}{9} = 1 \quad \checkmark$$

ii) $b^2 = a^2(e^2 - 1)$
 $9 = 4(e^2 - 1)$

$$e^2 = \frac{13}{4}$$

$$\therefore e = \frac{\sqrt{13}}{2} \quad \checkmark$$

iii) $ae = 2 \times \frac{\sqrt{13}}{2}$
 $= \sqrt{13}$

$\therefore$ foci: $(\sqrt{13}, 0)$ and $(-\sqrt{13}, 0)$ $\checkmark$

iv) $\frac{a}{e} = \frac{2}{\frac{\sqrt{13}}{2}}$
 $= \frac{4}{\sqrt{13}}$

$\therefore$ directrices: $x = \frac{4}{\sqrt{13}}$ or $x = -\frac{4}{\sqrt{13}}$ $\checkmark$

or $x = \pm \frac{4\sqrt{13}}{13}$

b) $P(x) = 3x^3 - 10x^2 + 7x + 10$

i) $\overline{2-i} = 2+i$ must also be a zero ✓

Let α be the third zero.

From sum of zeros:

$$\alpha + 2+i + 2-i = \frac{10}{3}$$

$$\therefore \alpha = -\frac{2}{3} \quad \checkmark$$

OR, From product of zeros:

$$\alpha(2+i)(2-i) = -\frac{10}{3}$$

$$5\alpha = -\frac{10}{3}$$

$$\therefore \alpha = -\frac{2}{3} \quad \checkmark$$

ii) $P(x) = 3\left(x + \frac{2}{3}\right)(x - (2-i))(x - (2+i)) \quad \checkmark$

$$= (3x+2)(x^2-4x+5) \quad \checkmark$$

c) i) Let $P(x) = 2x^3 - 9x^2 + 12x - 4$

$$P'(x) = 6x^2 - 18x + 12$$

$$= 6(x^2 - 3x + 2)$$

$$= 6(x-1)(x-2)$$

$$\therefore P'(x) = 0 \quad \text{when } x=1 \text{ or } x=2 \quad \checkmark$$

$$P(1) = 1 \quad \therefore x=1 \text{ is not the double root}$$

$$P(2) = 0 \quad \therefore x=2 \text{ is the double root}$$

$$\therefore \alpha = 2 \quad \checkmark$$

ii) Let β be the other root.

From sum of roots:

$$2 + 2 + \beta = \frac{9}{2}$$

$$\therefore \beta = \frac{1}{2}$$

$\therefore$ the remaining root is $x = \frac{1}{2}$ ✓

OR, From product of roots:

$$2 \times 2 \times \beta = \frac{4}{2}$$

$$\therefore \beta = \frac{1}{2}$$

d) i) $m=1, n=2$

$$P(cp, \frac{c}{p}) \rightarrow (x_1, y_1)$$

$$S(c\sqrt{2}, c\sqrt{2}) \rightarrow (x_2, y_2)$$

$$x_Q = \frac{mx_2 + nx_1}{m+n}$$

$$= \frac{1 \times c\sqrt{2} + 2 \times cp}{1+2}$$

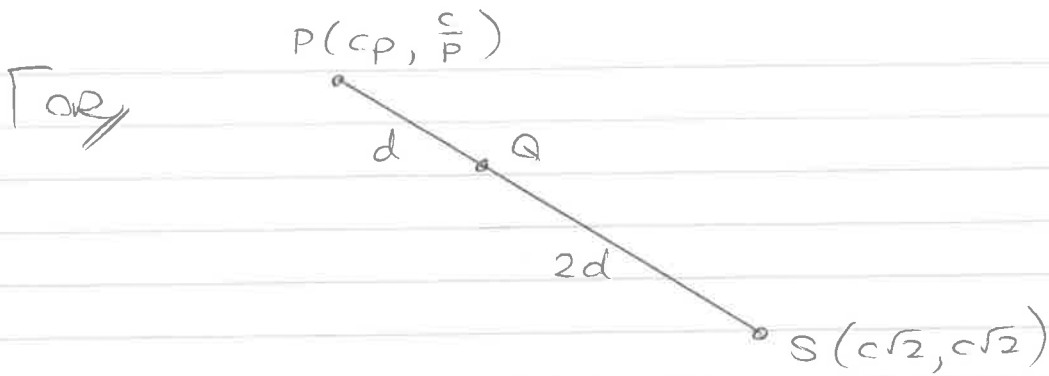
$$= \frac{c\sqrt{2} + 2cp}{3} = \frac{2cp + c\sqrt{2}}{3}$$
 ✓

$$y_Q = \frac{my_2 + ny_1}{m+n}$$

$$= \frac{1 \times c\sqrt{2} + 2 \times \frac{c}{p}}{1+2} \times \frac{p}{p}$$

$$= \frac{cp\sqrt{2} + 2c}{3p} = \frac{2c + cp\sqrt{2}}{3p}$$
 ✓

$$\therefore Q \left(\frac{2cp + c\sqrt{2}}{3}, \frac{2c + cp\sqrt{2}}{3p} \right)$$



$$\begin{aligned} x_Q &= x_P + \frac{x_S - x_P}{3} \\ &= cp + \frac{c\sqrt{2} - cp}{3} \\ &= \frac{2cp + c\sqrt{2}}{3} \end{aligned}$$

$$\begin{aligned} y_Q &= y_P - \frac{y_P - y_S}{3} \\ &= \frac{c}{p} - \frac{\frac{c}{p} - c\sqrt{2}}{3} \\ &= \frac{3c - (c - cp\sqrt{2})}{3p} \\ &= \frac{2c + cp\sqrt{2}}{3p} \end{aligned}$$

ii) $x = \frac{2cp + c\sqrt{2}}{3}$ (from (i))

$$p = \frac{3x - c\sqrt{2}}{2c} \quad \checkmark \text{ ————— } \textcircled{1}$$

$$y = \frac{2c + cp\sqrt{2}}{3p}$$

$$3yp - cp\sqrt{2} = 2c$$

$$p = \frac{2c}{3y - c\sqrt{2}} \quad \text{————— } \textcircled{2}$$

$$\textcircled{1} = \textcircled{2} :$$

$$\frac{3x - c\sqrt{2}}{2c} = \frac{2c}{3y - c\sqrt{2}}$$



$$(3x - c\sqrt{2})(3y - c\sqrt{2}) = 4c^2$$

QUESTION 12:

$$\begin{aligned} \text{a) } P(x) &= Q(x) \times D(x) + R(x) \\ &= Q(x)(x-2)(x-3) + ax + b \end{aligned} \quad \checkmark$$

$$P(2) = 4:$$

$$4 = 2a + b \quad (1)$$

$$P(3) = 3:$$

$$3 = 3a + b \quad (2) \quad \checkmark$$

$$(2) - (1):$$

$$a = -1$$

Sub into (1):

$$4 = -2 + b$$

$$b = 6$$

$\therefore$ the remainder is $-x + 6$ (or $6 - x$) $\checkmark$

$$\text{b) i) } \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = g$$

$$\frac{1}{2} v^2 = gx + C \quad \checkmark$$

when $x=0$, $v=0$:

$$0 = 0 + C \rightarrow C = 0$$

$$\therefore \frac{1}{2} v^2 = gx$$

$$v^2 = 2gx \quad \checkmark$$

(must show calc. of constant)

when $x=d$:

$$v^2 = 2gd.$$

$$\text{ii) } \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = g - k(x-d)$$

[notice that this equation is also valid when $x=d$]

$$\frac{1}{2} v^2 = gx - \frac{k(x-d)^2}{2} + C_1 \quad \checkmark$$

$$v^2 = 2gx - k(x-d)^2 + C_2$$

when $x = d$, $v^2 = 2gd$:

$$2gd = 2gd - k(d-d)^2 + C_2$$

$$\therefore C_2 = 0$$

$$\therefore v^2 = 2gx - k(x-d)^2 \text{ as required.}$$

c) $t = 0$
 $v = 7\sqrt{6} \text{ m/s}$
 $\ddot{x} = -(9.8 + 0.1v^2)$ $\uparrow^+$

i) $\frac{dv}{dt} = -(9.8 + 0.1v^2)$

$$\frac{dt}{dv} = -\frac{1}{9.8 + 0.1v^2}$$

$$t = -10 \int \frac{1}{98 + v^2} dv$$

$$= -\frac{10}{\sqrt{98}} \tan^{-1} \frac{v}{\sqrt{98}} + C$$

when $t = 0$, $v = 7\sqrt{6}$:

$$0 = -\frac{10}{\sqrt{98}} \tan^{-1} \sqrt{3} + C$$

$$\therefore C = \frac{10}{\sqrt{98}} \times \frac{\pi}{3}$$

$$= \frac{10\pi}{21\sqrt{2}}$$

$$\therefore t = -\frac{10}{\sqrt{2}} \tan^{-1} \frac{v}{\sqrt{2}} + \frac{10\pi}{21\sqrt{2}}$$

$$\left(\text{or } t = -\frac{5\sqrt{2}}{7} \tan^{-1} \frac{v}{\sqrt{2}} + \frac{5\pi\sqrt{2}}{21} \right)$$

ii) $t = ?$ when $v = 0$:

$$t = 0 + \frac{10\pi}{21\sqrt{2}}$$

$$= 1.06 \text{ seconds (to 3 sig. fig.)} \quad \checkmark$$

iii) $v \frac{dv}{dx} = -(9.8 + 0.1v^2)$

$$\frac{dx}{dv} = - \frac{v}{9.8 + 0.1v^2} \quad \checkmark$$

$$x = - \frac{10}{2} \int \frac{2v}{98 + v^2} dv$$

$$= -5 \ln(98 + v^2) + C \quad \checkmark$$

when $x = 0$, $v = 7\sqrt{6}$:

$$0 = -5 \ln(98 + 294) + C$$

$$\therefore C = 5 \ln 392$$

$$x = -5 \ln(98 + v^2) + 5 \ln 392$$

$$= 5 \ln \frac{392}{98 + v^2} \quad \checkmark$$

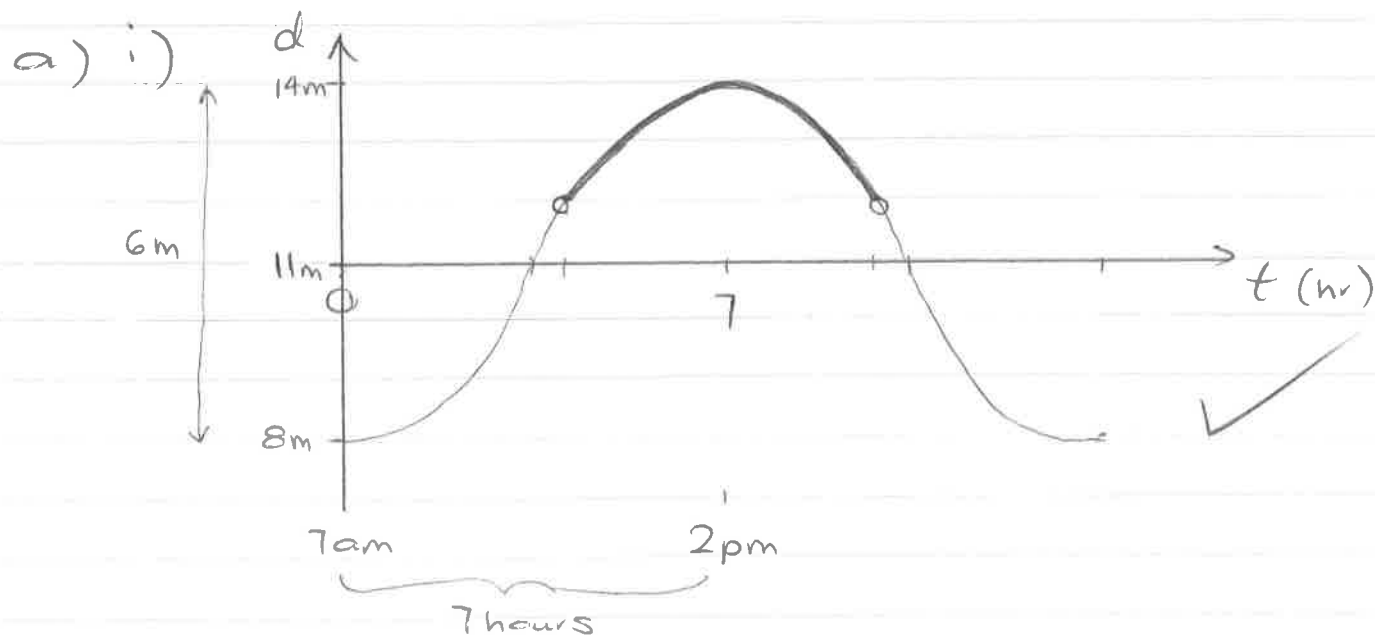
iv) $x = ?$ when $v = 0$:

$$x = 5 \ln \frac{392}{98}$$

$$= 5 \ln 4 \quad \checkmark$$

$$= 10 \ln 2 \text{ metres}$$

QUESTION 13:



$$a = 3$$

$$T = 2 \times 7 \\ = 14$$

$$T = \frac{2\pi}{n}$$

$$14 = \frac{2\pi}{n}$$

$$\therefore n = \frac{\pi}{7} \quad \checkmark$$

$$\therefore d = -3 \cos \frac{\pi}{7} t + 11 \quad \checkmark$$

ii) $12 = -3 \cos \frac{\pi}{7} t + 11$

$$\cos \frac{\pi}{7} t = -\frac{1}{3} \quad \checkmark$$

$$\frac{\pi}{7} t = \pi - \cos^{-1}\left(\frac{1}{3}\right) \quad \text{OR} \quad \pi + \cos^{-1}\left(\frac{1}{3}\right)$$

$$\therefore t = \frac{7}{\pi} (\pi - \cos^{-1}(\frac{1}{3})) \quad \text{OR} \quad \frac{7}{\pi} (\pi + \cos^{-1}(\frac{1}{3}))$$

$$= 4.2572 \dots \quad \checkmark$$

$$\doteq 4 \text{ hr } 15 \text{ min}$$

$$= 9.7427 \dots$$

$$\doteq 9 \text{ hr } 45 \text{ min}$$

$\therefore$ it can pass through between 11:15am + 4:45pm

$$b) i) \alpha\beta\gamma = -\frac{(-2)}{2}$$

$$= 1$$



$$ii) \frac{\alpha\beta}{\gamma} = \frac{\alpha\beta\gamma}{\gamma^2}$$

$$= \frac{1}{\gamma^2}$$

$$\frac{\alpha\gamma}{\beta} = \frac{\alpha\beta\gamma}{\beta^2}$$

$$= \frac{1}{\beta^2}$$

$$\frac{\beta\gamma}{\alpha} = \frac{\alpha\beta\gamma}{\alpha^2}$$

$$= \frac{1}{\alpha^2}$$

$\therefore$ the required roots are $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$



$\therefore$ replace x with $\frac{1}{\sqrt{x}}$:

$$2\left(\frac{1}{\sqrt{x}}\right)^3 - 9\left(\frac{1}{\sqrt{x}}\right)^2 + 8\left(\frac{1}{\sqrt{x}}\right) - 2 = 0$$



$$2 - 9\sqrt{x} + 8x - 2x\sqrt{x} = 0$$

$$(2 + 8x)^2 = [\sqrt{x}(2x + 9)]^2$$

$$4 + 32x + 64x^2 = 4x^3 + 36x^2 + 81x$$

$$4x^3 - 28x^2 + 49x - 4 = 0$$



c) i) $y = x^3 - 3ax + b$

$$\frac{dy}{dx} = 3x^2 - 3a$$

$$\frac{d^2y}{dx^2} = 6x$$

St. points:

$$\frac{dy}{dx} = 0 \quad \text{when} \quad 3x^2 = 3a$$

$$x = \pm\sqrt{a} \quad \checkmark$$

When $x = \sqrt{a}$, $y = a\sqrt{a} - 3a\sqrt{a} + b$
 $= -2a\sqrt{a} + b$

$$\frac{d^2y}{dx^2} = 6\sqrt{a}$$

$> 0 \therefore$ minimum turning point
 at $(\sqrt{a}, -2a\sqrt{a} + b) \quad \checkmark$

When $x = -\sqrt{a}$ $y = -a\sqrt{a} + 3a\sqrt{a} + b$
 $= 2a\sqrt{a} + b$

$$\frac{d^2y}{dx^2} = -6\sqrt{a}$$

$< 0 \therefore$ maximum turning point
 at $(-\sqrt{a}, 2a\sqrt{a} + b) \quad \checkmark$

ii) 3 distinct real roots

$\therefore$ turning points must be on opposite sides of the x -axis. $\checkmark$

$$\therefore y_{\max} \times y_{\min} < 0$$

$$(2a\sqrt{a} + b)(-2a\sqrt{a} + b) < 0 \quad \checkmark$$

$$b^2 - 4a^3 < 0$$

$$b^2 < 4a^3$$

QUESTION 14:

$$a) I_n = \int_0^1 \frac{1}{(x^2+1)^n} dx$$

$$i) I_n = \int_0^1 \underbrace{\frac{d}{dx}(x)} \times \underbrace{(x^2+1)^{-n}} dx \quad \checkmark$$

$$= \left[x(x^2+1)^{-n} \right]_0^1 - \int_0^1 x \times -2nx(x^2+1)^{-n-1} dx$$

$$= 2^{-n} + 2n \int_0^1 \frac{x^2}{(x^2+1)^{n+1}} dx \quad \checkmark$$

$$= 2^{-n} + 2n \int_0^1 \frac{x^2+1-1}{(x^2+1)^{n+1}} dx$$

$$= 2^{-n} + 2n \int_0^1 \left(\frac{1}{(x^2+1)^n} - \frac{1}{(x^2+1)^{n+1}} \right) dx \quad \checkmark$$

$$= 2^{-n} + 2n [I_n - I_{n+1}]$$

$$2n I_{n+1} = 2^{-n} + 2n I_n - I_n \quad \checkmark$$

$$\therefore I_{n+1} = \frac{1}{2n} [2^{-n} + (2n-1) I_n]$$

$$ii) I_1 = \int_0^1 \frac{1}{x^2+1} dx$$

$$= [\tan^{-1} x]_0^1$$

$$= \frac{\pi}{4} \quad \checkmark$$

$$I_3 = \frac{1}{4} [2^{-2} + 3I_2]$$

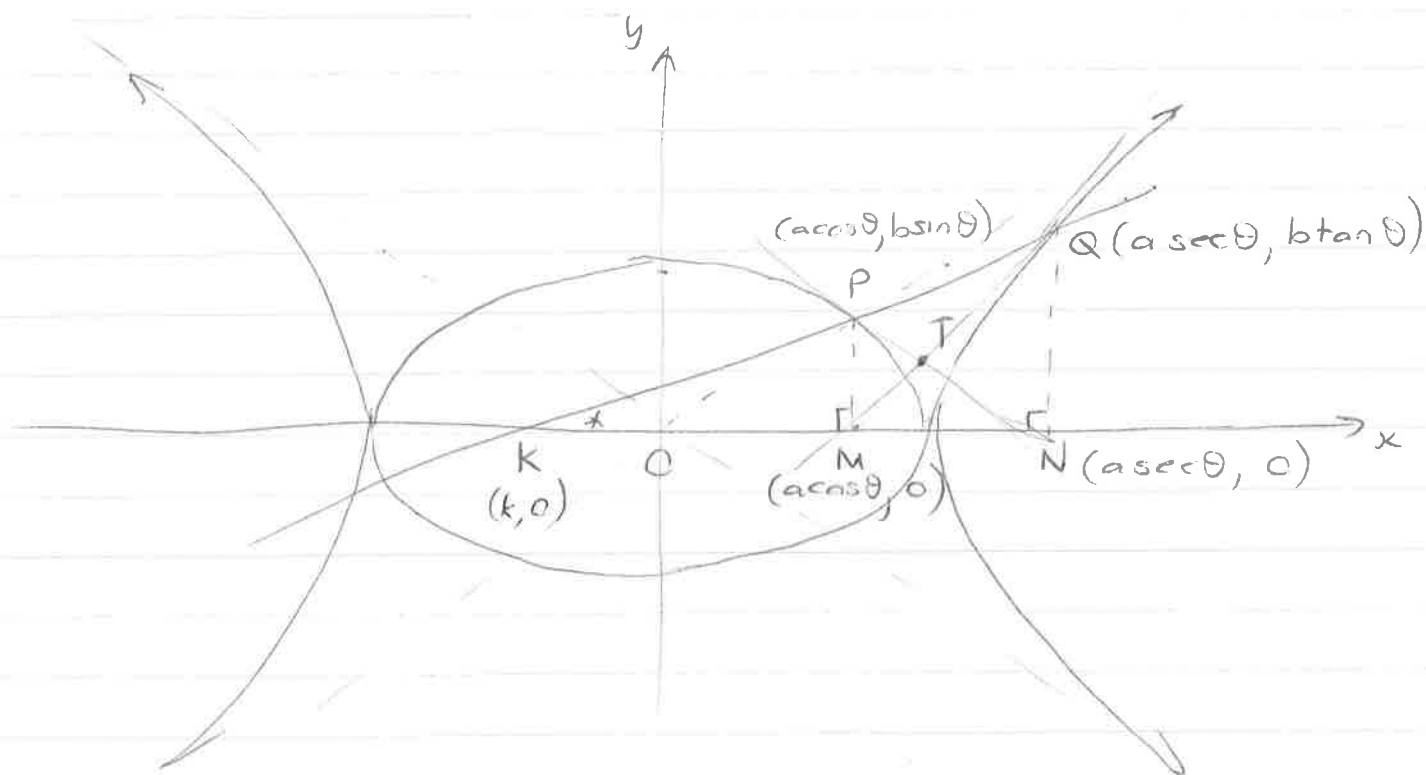
$$= \frac{1}{16} + \frac{3}{4} \left[\frac{1}{2} (2^{-1} + I_1) \right]$$

$$= \frac{1}{16} + \frac{3}{16} + \frac{3}{8} \times \frac{\pi}{4}$$

$$= \frac{8 + 3\pi}{32} \quad \checkmark$$

b)

NTS



b) i) Clearly $\triangle KPM \parallel \triangle KQN$ (equiangular)

$$\begin{aligned} \text{So } \frac{KM}{KN} &= \frac{PM}{QN} \quad (\text{matching side in similar } \triangle\text{'s}) \\ &= \frac{b \sin \theta}{b \tan \theta} \quad \checkmark \\ &= \cos \theta \end{aligned}$$

ii) Let $K = (k, 0)$, then:

$$\begin{aligned} KM &= KN \cos \theta \\ a \cos \theta - k &= (a \sec \theta - k) \cos \theta \quad \checkmark \\ a \cos \theta - k &= a - k \cos \theta \\ a(\cos \theta - 1) &= k(1 - \cos \theta) \end{aligned}$$

$$\begin{aligned} \therefore k &= -a \\ \therefore K &(-a, 0) \quad \checkmark \end{aligned}$$

iii) Gradient of tangent at P:

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \\ &= \frac{b \cos \theta}{-a \sin \theta} \quad \checkmark \end{aligned}$$

Point-gradient formula:

$$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$$

$$\frac{bx \cos \theta}{ab} + \frac{ay \sin \theta}{ab} = \frac{ab}{ab} (\cos^2 \theta + \sin^2 \theta) \quad \checkmark$$

$$\therefore \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

When $y = 0$:

$$\frac{x \cos \theta}{a} = 1$$

$$\therefore x = a \sec \theta \quad \checkmark$$

$\therefore$ the tangent passes through $N(a \sec \theta, 0)$

$$\begin{aligned} \text{iv) } \alpha) \quad \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} &= 1 & \text{--- (1)} \\ \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} &= 1 & \text{--- (2)} \end{aligned} \quad \left. \vphantom{\begin{aligned} \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} &= 1 \\ \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} &= 1 \end{aligned}} \right\} \begin{array}{l} \text{from given} \\ \text{information} \end{array}$$

$$\textcircled{1} \div \cos \theta: \quad \frac{x}{a} + \frac{y \tan \theta}{b} = \sec \theta \quad \text{--- } \textcircled{1}^*$$

$$\textcircled{1}^* + \textcircled{2}: \quad \frac{x}{a} (1 + \sec \theta) = \sec \theta + 1$$

$$\frac{x}{a} = 1$$

$$x = a$$

sub into $\textcircled{1}$:

$$\cos \theta + \frac{y \sin \theta}{b} = 1$$

$$y = \frac{b(1 - \cos \theta)}{\sin \theta}$$

$$\left(\text{or } \frac{b \sin \theta}{1 + \cos \theta} \right)$$

$$\therefore T \left(a, \frac{b(1 - \cos \theta)}{\sin \theta} \right)$$

so T always lies on the line $x = a$. $\checkmark$

$$\beta) \quad y = \frac{b(1 - \cos \theta)}{\sin \theta}$$

$$= \frac{b \left(1 - \frac{1-t^2}{1+t^2}\right)}{\frac{2t}{1+t^2}}$$

$$= \frac{b(1+t^2 - (1-t^2))}{2t}$$

$$= \frac{b \times 2t^2}{2t}$$

$$= bt$$

$$= b \tan \frac{\theta}{2}$$

$$\text{Let } t = \tan \frac{\theta}{2}$$

$$\therefore \sin \theta = \frac{2t}{1+t^2}$$

$$\cos \theta = \frac{1-t^2}{1+t^2}$$

$$-1 < \tan \frac{\theta}{2} < 1 \quad \text{for } -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$$\therefore -b < b \tan \frac{\theta}{2} < b$$

$\therefore T$ lies on the line $x=a$
where $-b < y < b$ ✓

BUT $y \neq 0$ since $P \neq Q$ are distinct points $\therefore$ the range of y -values for T should exclude $y=0$

$\gamma)$ The algebra above does not change,
so T is still on the line $x=a$

$$\text{for } \frac{\pi}{2} < \theta < \pi, \quad b < b \tan \frac{\theta}{2} < \infty$$

$$\neq \text{for } -\pi < \theta < -\frac{\pi}{2}, \quad -\infty < b \tan \frac{\theta}{2} < -b$$

$\therefore T$ lies on the line $x=a$
where $y < -b$ or $y > b$ ✓